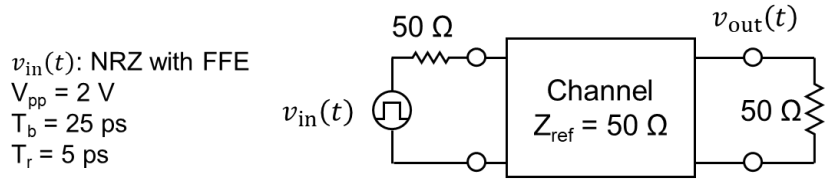
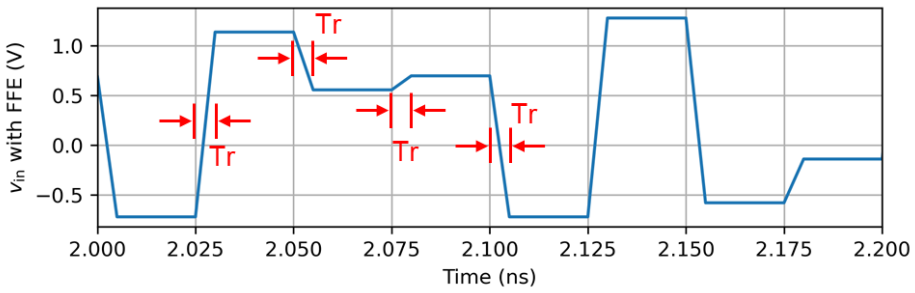
Homework: FFE simulation and design

In the previous homework, we design a CTLE for a 50cm strip line channel. In this homework, we are going to design a Tx FFE for the same channel (so please use the same s2p). Same parameters as before (40 Gbps NRZ, risetime 5 ps, impedance matching, and that the transfer function from $V_{\text{in}}(f)$ to $V_{\text{out}}(f)$ is $H = S_{21}^{\text{Ch}}/2$). However, we are not going to use the frequency response of the FFE to compute the output waveform. Instead, we will construct the time-domain post-FFE waveform explicitly and then do sectional convolution. The frequency response of FFE is only for reference.



Note that the same risetime is used for all kinds of transition:



Use the zero-forcing method to compute the tap values. Show the resulting eye diagrams of

1. Unequalized eye (same as previous homework)
2. One precursor, no post cursors.
3. One post cursor, no precursors.
4. One precursor and one post cursor.
5. One precursor and two post cursors.

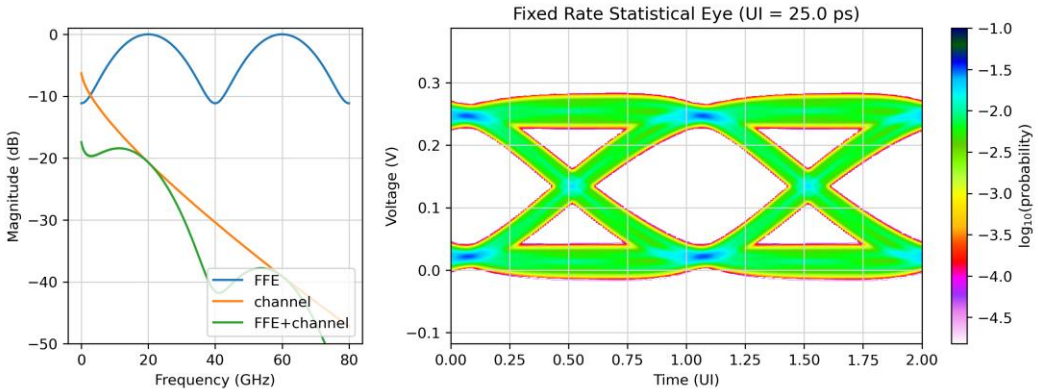
Also, show two snapshots of the post-FFE waveform (at Tx) and post-channel waveform (at Rx) of Case 4 (one precursor and one post cursor). Total four snapshots.

Optional problem:
Implement your FFE in ADS ChannelSim to test its performance. See appendix.

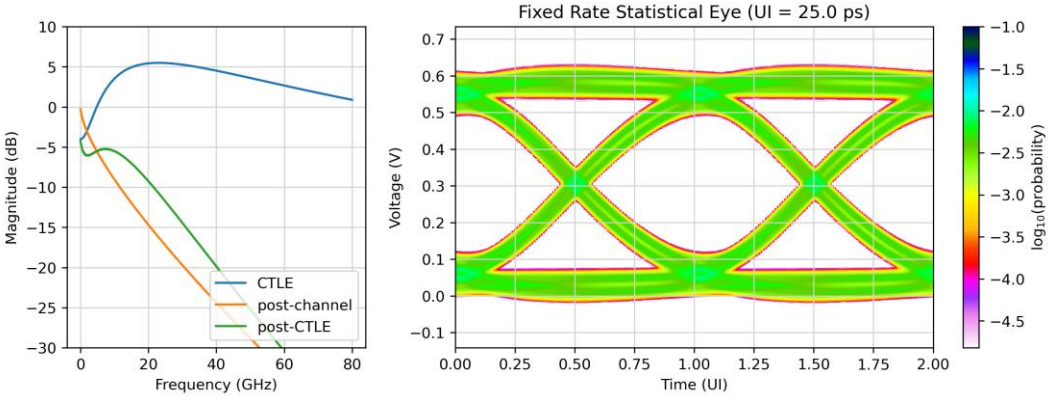
Note: In this homework, the Python code for doing sectional convolution is different from the previous homework in how the input waveform is constructed. To build a waveform with FFE, we will first perform discrete convolution of the input binary signal and the FFE tap vector. Then, the exact time-domain waveform including finite risetime is constructed based on such discrete input data.

Appendix 1: Example answer

Eye with three-tap FFE (one precursor and one post-cursor):

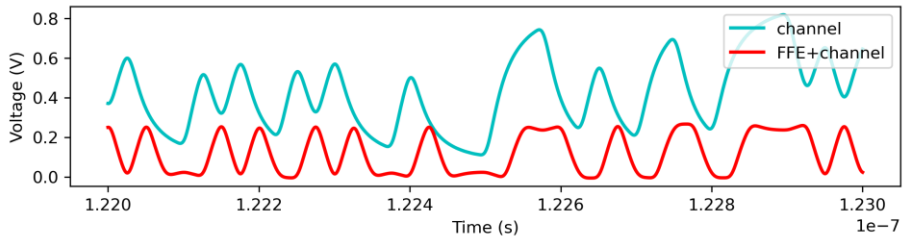


You can compare with the eye with CTLE in the previous homework:

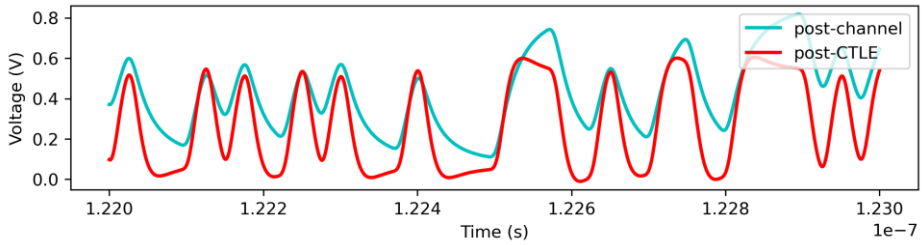


What difference do you observe?

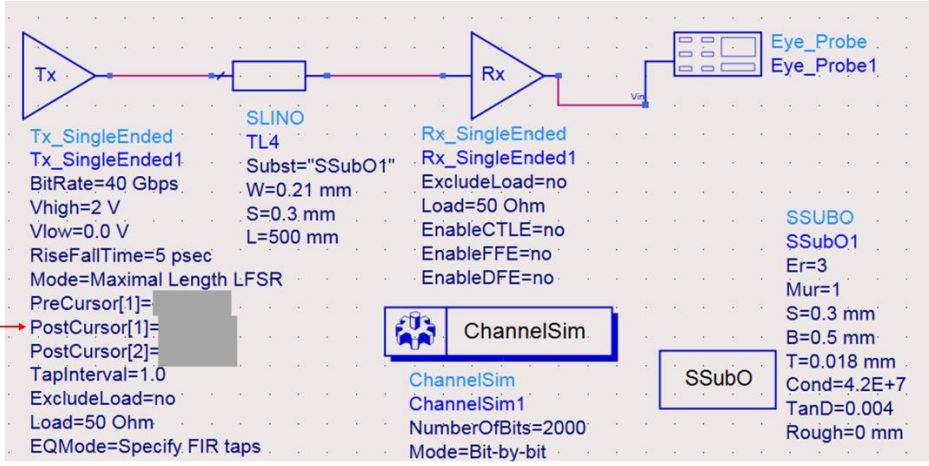
Example of post-FFE waveform is shown in previous page. Example of post-channel waveform:



Again, compared with the post-CTLE waveform from the previous homework, what do you observe?

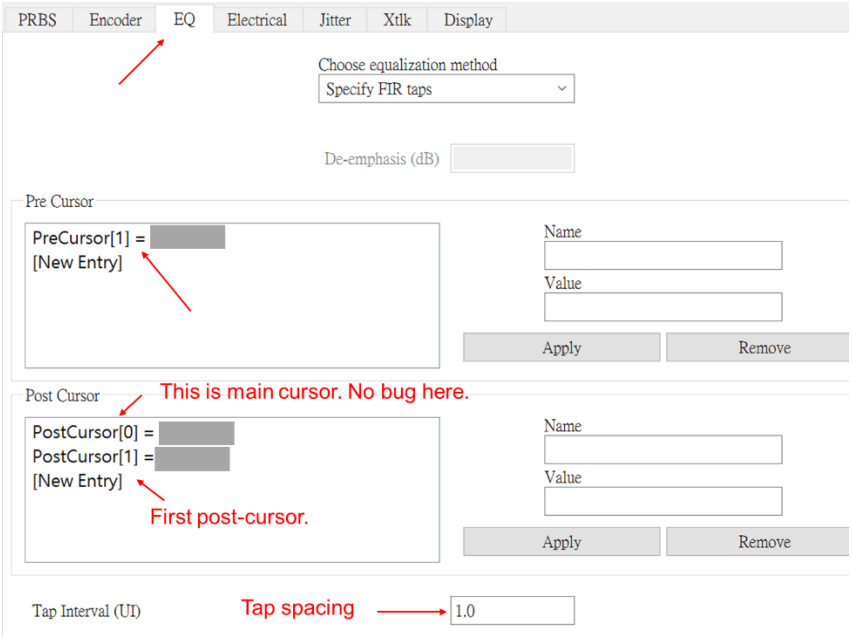


Appendix 2: ADS setting



This means main cursor. Should be PostCursor[0]. It's a bug of ADS [version 2021].

Double click the Tx to set the FFE:



ADS's eye and post-channel waveform (with FFE):

